INCH-POUND

MIL-I-5154B(USAF)
AMENDMENT 4

30 January 1989
SUPERSEDING
AMENDMENT 3
6 July 1983

MILITARY SPECIFICATION

INDICATOR, TRUE AIRSPEED, TYPE-M-5

This amendment forms a part of MIL-I-5154B, dated 16 January 1968, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 Delete "first sentence" and add: "Government documents."

* Add: "2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)."

* Under "SPECIFICATIONS, Federal," delete "QQ-P-416"; add the following: "PPP-B-601 Boxes, Wood, Cleated Plywood" and "PPP-B-636 Box, Shipping, Fiberboard."

* Under SPECIFICATIONS, Military", delete "MIL-E-5272" and add: "MIL-P-26514 Polyurethane Foam, Rigid or Flexible, for Packaging" and "MIL-C-83488 Coating, Aluminum, ION Vapor Deposited."

Add the following boxed paragraph to bottom center of page:

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

Under "STANDARDS, Military," delete "MIL-STD-143", "MIL-STD-726", "MIL-STD-794", "MIL-STD-834"and add: "MIL-STD-810C Environmental Test Methods and Engineering Guidelines", Order of Preference For the Selection Of" and "MIL-STD-2073-1 DOD Material Procedures For Development And Application Of Packaging Requirements."

* Delete the "parenthetical source statement that follows the listing of Government specifications and standards" and add: "(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)"

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil - Downloaded: 2026-04-17 04:23Z
Check the source to verify that this is the current version before use.

* "(Copies of the MIL-STD-810C is available from Director Naval Publications and Printing Service Office, 700 Robbins Ave, Bldg 4 Section D, Philadelphia PA 19111.)"

* Add: "2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)."

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be address to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* Add: "2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

* Delete "3.1" entirely and add: "3.1 First article. When specified° (see 6.4), a sample shall be subjected to first article inspection in accordance with 4.3."

* 3.2 Line 3 delete "MIL-STD-143" and add: "MIL-STD-970."

PAGE 10

* Delete "3.7.2" entirely and add: "3.7.2 "Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practical, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating in lieu of MIL-C-83488 may be used if demonstrated to be satisfactory and approved by preparing activity. Cadmium plating must be avoided when satisfactory alternative process can be used."

* Add: "3.11 Recycled and reclaimed material. Recycled and reclaimed material shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

PAGE 11

Delete "4.1" entirely and add: "4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed

Source: https://assist.dla.mil -- Downloaded: 202610-17T04:23Z
Check the source to verify that this is the current version before use.

necessary to ensure supplies and services conform to prescribed requirments."

* Add: "4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material."

* Delete "4.2" and "a" entirely and add new: "4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)."

Delete "4.3" entirely and add new: "4.3 First article inspection. First article shall consist of all the tests of this specification."

Delete "4.3.1" entirely and add new: "4.3.1 First article samples. First article inspection shall be performed on three indicators."

PAGE 12

Delete "4.5 Test conditions" and add: "4.5 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test conditions specified in paragraph 4.5.1 in this specification.

PAGE 17

4.6.2 Line 2 delete "MIL-E-5272" and add: "MIL-STD-810C."

4.6.2.1 Delete "Procedure II" and add: "Procedure I of Method 502.1 of MIL-STD-810C." Also, delete "-54°C" and add: "-57°C".

4.6.2.2 Delete "Procedure II" and add: "Procedure I of Method 501.1 of MIL-STD-810C."

4.6.2.3 Delete paragraph entirely and add the following: "4.6.2.3 Temperature-altitude. The indicator shall be installed in a test chamber in accordance with MIL-STD-810C in a manner as similar to that employed in service use as possible.

If the equipment requires cooling air from a source other than the compartment ambient, appropriate ducting shall be installed to provide air in volume, temperature range, and density as necessary. The chamber shall be maintained at an altitude of 60,000 feet and at an ambient temperature of 54°C for a continuous period of four hours. The selected combination of conditions shall be maintained and the equipment operated for a sufficient length of time to insure thermal stabilization before operational checks are made. Operational checks shall be made in like manner, in any desired sequence, at a sufficient number of succeeding check points to adequately portray operational capabilities of the equipment. When testing within limits of the continuous operating capabilities of the equipment,

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

the equipment may continue to operate while chamber conditions are being changed from one test point to another, in order to reduce time required to attain thermal stability. For intermittent and short time operation, the temperature-altitude limits, exposure time, duty cycle and operational checks shall be as required by the applicable equipment specification."

4.6.2.4 Delete entirely and add the following: "4.6.2.4 Vibration. The procedure of paragraph 4.6.2.4.1 shall be used, modified as follows. The test shall be conducted while a differential pressure of 300 knots is applied to the indicator. The difference between the indication of the pointer at the beginning and at the end of the three hour vibration period shall not exceed the tolerance specified in Table VII, and no looseness in the mechanism nor damage shall result in any part of the indicator."

Add: "4.6.2.4.1 Constructional details. This procedure constitutes a vibration failure test to detect faulty constructional details in panel-mounted aircraft instruments of reciprocating engine type aircraft. The indicator shall be mounted on the apparatus in its normal operating position. While being operated in accordance with the detail specification, the instrument shall be vibrated with circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45 degrees to the horizontal plane and the frequency of vibration shall be varied uniformly from 5 to 50 cycles per second and return once each hour for a 3 hour period.

PAGE 18

4.6.2.5 Delete "Procedure I" and add: "Method 508.2 of MIL-STD-810C".

4.6.2.6 Delete "Procedure I for a period of 50 hours" and add "Procedure I of Method 509.1 of MIL-STD-810C."

4.6.2.7 Delete "Procedure I" and add: "Procedure I of Method 509.1 of MIL-STD-810C."

PAGE 22

4.6.15 Delete the first sentence and add: "Vibration error shall be in accordance with 4.6.15.1."

Add new paragraph as follows: "4.6.15.1 Vibration errors. This procedure applies to the determination of vibration errors in panel-mounted aircraft instruments of reciprocating engine type aircraft.

The indicator shall be mounted on the apparatus in its normal operating position. While being operated, the item shall be vibrated with circular motion in a plane inclined 45 degrees to the horizontal plane with an amplitude between 0.009 to 0.011 inch. The frequency of applied vibration shall be varied slowly from 5 to 50 cycles per second. No condition of applied vibration shall cause vibration errors involving pointer oscillation or variation in excess of the tolerances specified herein."

PAGE 23

* 6.2 Delete "title and first sentence" and add: "Acquisition requirements. Acquisition documents must specify the following:"

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

* Add: "6.2.c Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)."

Delete Section 5 in its entirely and add:

* 5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (See 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 or MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method 2M of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The industrial preservation of indicators shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or industrial as specified (see 6.2).

5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contractor or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6.2,b Delete and add: "b. Selection of applicable levels of packaging and packing (see section 5)."

Add: "6.2,c Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1)."

PAGE 24

6.4. Delete entirely and all: "6.4 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition document regarding arrangements for examinations, test and approval of the first article.

* Delete "6.5".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 71

Project number:
6610-F280

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.